

Automobile Leasing : A Markov Process

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Introduction

Markov Chains have a wide range of applications, from finance and economics to natural language processing and speech recognition.

Here we will study an important application of Markov Chain in real life.

Markov Process

Stochastic Process

Any sequence of experiments whose outcome at any stage depends on chance

Markov Process

Stochastic Process with the following properties:

- I. finite outcomes.
- II. Dependence of Probability only on previous outcome.
- III. Constant probability over time.

Automobile Leasing

Automobile Dealer leases four types of vehicles:

- I. Four Door Sedans.
- II. Sports Cars.
- III. Minivans.
- IV. Sports utility vehicles.

After 2 years customers have to renegotiate the lease and choose a new vehicle.

Automobile Leasing (Continued)

Table: Transition Probabilities for Vehicle Leasing

Current Lease				
Sedan	Sports Car	Minivan	SUV	Next Lease
0.80	0.10	0.05	0.05	Sedan
0.10	0.80	0.05	0.05	Sports Car
0.05	0.05	0.80	0.10	Minivan
0.05	0.05	0.10	0.80	SUV

Automobile Leasing (Continued)

Working Example

Suppose, We have initially:

I. 200 Sedans.

II. 100 Sports Cars.

III. 100 Minivans.

IV. 100 SUVs.

Set the transition matrix,

$$A = \begin{bmatrix} 0.80 & 0.10 & 0.05 & 0.05 \\ 0.10 & 0.80 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.80 & 0.10 \\ 0.05 & 0.05 & 0.10 & 0.80 \end{bmatrix}$$

$$\& x_0 = \begin{bmatrix} 200 \\ 100 \\ 100 \\ 100 \end{bmatrix}$$

Automobile Leasing (Continued)

$$x_1 = Ax_0 = \begin{bmatrix} 180 \\ 110 \\ 105 \\ 105 \end{bmatrix}$$

Consequently, for future leases $x_{n+1} = Ax_n$ for $n = 1, 2, \dots$

Here x_i = State Vectors for $i = 1, 2, \dots$

Markov Chain

Sequence of the State Vectors.

Formation of Stochastic Matrix

Stochastic Matrix

Entries are non-negative and entries in each column add up to 1.

Divide entries of initial vector by 500. Then entries of the new initial state

$$\text{vector } x_0 = \begin{bmatrix} 0.40 \\ 0.20 \\ 0.20 \\ 0.20 \end{bmatrix} .$$

Then succeeding state vectors of the chain will be probability vectors.

Long Range Behaviour

Long Range Behaviour of the process is determined by Eigen values and Eigen vectors of matrix A .

Eigen values of A : $\lambda_1 = 1, \lambda_2 = 0.8, \lambda_3 = \lambda_4 = 0.7$

A has four linearly independent Eigenvectors. Hence Diagonalization is possible.

Use Eigen Vectors to form a diagonalizing matrix Y . Then

$$A = YDY^{-1} = \begin{bmatrix} 1 & -1 & 0 & 1 \\ 1 & -1 & 0 & -1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{8}{10} & 0 & 0 \\ 0 & 0 & \frac{7}{10} & 0 \\ 0 & 0 & 0 & \frac{7}{10} \end{bmatrix} \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 \end{bmatrix}$$

Convergence of Probability Matrix

set the state vectors,

$$\begin{aligned}x_n &= YD^nY^{-1}x_0 \\&= YD^n(0.25, -0.05, 0, 0.10)^T \\&= Y(0.25, -0.05(0.8)^n, 0, 0.10(0.7)^n)^T \\&= 0.25 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - 0.05(0.8)^n \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix} + 0.10(0.7)^n \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}\end{aligned}$$

As n increases, x_n approaches the steady-state vector

$$x = (0.25, 0.25, 0.25, 0.25)^T$$

Convergence of Probability Matrix (Continued)

Prediction

Leases will be divided equally among four types of vehicles.

Theorem

If $\lambda_1 = 1$ is a dominant Eigenvalue of a Stochastic Matrix A , then the Markov Chain with transition A will converge to a steady-state vector.

Convergence of Probability Matrix (Continued)

Defective Matrix

An $n \times n$ matrix having fewer than n linearly independent Eigenvectors.

Convergence is still possible by Jordan canonical form.

Convergence is unique for a matrix having positive entries.

References

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